

A Novel Exchange Rate Forecasting Paradigm Based on Multi-Agent Collaboration and Multimodal Big Data-Driven Methods

Di Han, Jiaqi Chen*, Zikun Guo, Qixian Li, and Bocheng Wang

Abstract: The existing methodologies for forecasting exchange rates exhibit significant accuracy, systematicity, and efficiency deficiencies. Traditional exchange rate forecasting methods are limited by siloed data acquisition, cumbersome data processing, subjective feature engineering, and low efficiency model forecasting. To address these limitations, this paper proposes an exchange rate forecasting paradigm based on multimodal big data. The paradigm effectively addresses the challenges of non-standardized feature selection inherent in traditional forecasting methods by introducing flexible algorithmic mechanisms and dynamic feature selection strategies. On this basis, this paper proposes the FX-Agents framework, which enables efficient data acquisition and processing through agent collaboration driven by large language models (LLMs). Its flexible multi-agent module design ensures efficient and stable forecasting performance. Experimental results demonstrate that FX-Agents outperforms traditional methods in forecasting accuracy and processing efficiency. The source code of our work is publicly available at <https://github.com/Kon-Kwok/FX-Agent>.

Key words: multi-agent; paradigm; multimodal big data; exchange rate forecasting

1 Introduction

The growing volatility of the CNY exchange rate presents challenges for accurate and robust forecasting. Current forecasting approaches face four main limitations. Firstly, traditional models, such as generalized autoregressive conditional heteroskedasticity (GARCH), linear models, and autoregressive integrated moving average (ARIMA),

often rely on single-source or limited-scope data, leading to incomplete data usage and hindering the comprehensive integration of diverse and relevant information^[1-3]. Secondly, the data processing procedures are manual, complex, and inefficient, lacking standardization and automation^[4]. Thirdly, feature selection experience cannot be effectively consolidated, and the heavy reliance on expert knowledge not only limits the adaptability and generalizability of the model, but also introduces subjective biases and instability. Finally, the forecasting stage relies heavily on expert intervention for model selection and hyperparameter tuning, resulting in a subjective and inefficient process. Moreover, traditional methods often lack the integration of automated optimization mechanisms into financial forecasting workflows and agent-based systems, which further limits their adaptability and deployment efficiency^[5].

Recent studies have attained results by utilizing deep learning techniques to analyze complex data. For

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instance, Han et al.^[6] and Liu et al.^[7] combined convolutional neural networks (CNNs) with long short-term memory (LSTM) models to extract significant features for forecasting. Similarly, Lopez-Lira and Tang^[8] employed large language models (LLMs) in financial forecasting, demonstrating that ChatGPT exhibits considerable capability in predicting market trends. However, these approaches remain limited in scope, often addressing individual parts of the forecasting process without offering a systematic approach to data processing and feature selection. Agent technology has emerged as a promising direction to address these limitations. Xi et al.^[9] showed that agents utilizing LLMs exhibit significant advantages in executing complex tasks. MetaGPT, a multi-agent system developed by Hong et al.^[10], further demonstrates the potential of agent-based systems to enhance collaborative efficiency. This shift from the local optimization of single parts to multi-agent collaboration enables coordinated handling of data acquisition, processing, feature extraction, and forecasting, thereby reducing reliance on expert intervention and enhancing standardization and automation^[11]. These existing works provide valuable insights for the development of exchange rate forecasting paradigms. Based on the above research, this paper proposes a novel exchange rate forecasting paradigm based on multi-agent collaboration and a multimodal big data-driven method. On this basis, we develop a corresponding multi-agent framework for exchange rate forecasting (denoted as FX-Agents). By integrating LLMs with multiple specialized agents, the FX-Agents framework transforms traditional forecasting into a fully agent-driven operational pipeline that automates comprehensive data acquisition, intelligent processing, knowledge-guided feature selection, and forecasting.

The main contributions of this paper are as follows:

(1) Multi-agent forecasting paradigm: This paper introduces an exchange rate forecasting paradigm based on multimodal big data. By incorporating flexible algorithmic mechanisms and dynamic feature selection strategies, the paradigm effectively addresses the non-standardized feature selection challenges inherent in traditional forecasting methods, thereby achieving efficient and stable forecasting performance.

(2) FX-Agents framework: Based on the above paradigm, this paper designs the FX-Agents framework, which achieves efficient data processing

through agent collaboration driven by LLMs, and utilizes a flexible multi-agent module design to ensure efficient and stable forecasting performance. This significantly improves overall operational efficiency in big data processing. Experimental results demonstrate that FX-Agents outperforms traditional exchange rate forecasting methods in accuracy and processing efficiency.

2 Related Work

Forecasting exchange rates has been extensively examined utilizing both statistical models and machine learning techniques. Traditional statistical methods, such as threshold autoregression^[12], differenced ARIMA models^[13], and GARCH models^[14], are capable of capturing linear dynamics and volatility, but they struggle to accommodate nonlinear relationships and offer limited generalization. In contrast, the rise of deep learning models has opened up new possibilities for modeling complex temporal patterns in exchange rates without strict assumptions. Recent works have shown encouraging results using architectures, such as CNN^[15], LSTM^[16], and Transformer-based models^[17]. Despite these advancements, most existing works still rely on unimodal inputs, such as historical exchange rate data or selected macroeconomic indicators. These datasets are typically low in dimensionality, updated infrequently, and fail to capture the real-time complexity of financial markets. In comparison, multimodal big data provide a promising approach to addressing these limitations and improving the practical accuracy of exchange rate forecasting models.

To address the lack of multimodal support, some researchers have attempted to manually incorporate unstructured data. Han et al.^[6] improved forecasting performance by introducing news sentiment derived from custom sentiment dictionaries. More recently, the emergence of LLMs, such as ChatGPT, has further advanced the field by facilitating multimodal data fusion and improving model interpretability. LLMs have shown the potential to overcome limitations in both statistical models and machine learning models by enabling richer semantic understanding and more flexible data integration. Lopez-Lira and Tang^[8] demonstrated the feasibility of LLMs in predicting stock market returns. In the context of exchange rate forecasting, Ullah and Huang^[18] and Han et al.^[6] applied LLMs and prompted engineering for sentiment analysis of financial news, improving accuracy through

the integration of multimodal big data. However, current LLM-based frameworks lack standardized output mechanisms, suffer from subjective feature selection, and are prone to hallucinations, all of which hinder reliability and generalizability. Moreover, both statistical and machine learning models continue to rely heavily on expert knowledge for data preprocessing and feature engineering, limiting the automation and scalability of the forecasting process.

Agent technologies offer a promising pathway to address these issues. In the financial domain, researchers have begun leveraging agents to enhance the reasoning capabilities of LLMs. FinGPT^[19] introduces an open-source framework for building financial LLMs. FinLlama^[20] demonstrates the effectiveness of fine-tuning models for specific tasks like sentiment analysis. FINMEM^[21] pioneers a cognitive agent with a human-like memory structure, while FinVerse^[22] showcases the power of a versatile agent with extensive tool-use capabilities. These studies highlight agent-based systems' growing potential for multi-source integration and robust decision-making in finance. To clarify the differences of our work among these diverse paradigms, Table 1 provides a detailed comparison. While each framework offers unique strengths, FX-Agents is distinguished by its structured, evidence-driven approach specifically tailored for exchange rate forecasting tasks.

As the comparison in Table 1 illustrates, a critical gap remains in the existing landscape. While current frameworks excel in areas like flexibility (FinVerse^[22]) or cognitive simulation (FINMEM^[21]), they do not offer an end-to-end, automated pipeline that explicitly addresses exchange rate forecasting tasks. To fill this gap, this paper proposes FX-Agents. Our framework is uniquely designed not only to automate the entire forecasting workflow through a multi-agent pipeline but also to ensure the reliability of its core decision-making process. By integrating a retrieval augmented generation (RAG) mechanism for proactive and evidence-based feature validation, FX-Agents introduces a new level of trustworthiness and auditability to automated financial forecasting, a contribution detailed in the next section.

3 FX-Agents: A New Forecasting Paradigm

This section first deconstructs the traditional forecasting process to identify its inherent limitations. Based on this analysis, it then introduces FX-Agents, a novel multi-agent paradigm developed to systematically overcome these challenges through coordinated, intelligent automation.

3.1 Limitations of traditional forecasting

The traditional exchange rate forecasting process is decomposed into four stages: perception, planning,

Table 1 Detailed comparison of financial LLM and agent frameworks.

Agent framework	Dimension				
	Core paradigm	Primary goal	Core mechanism	Key strength	Hallucination solution
FX-Agents (ours)	Multi-agent collaborative pipeline	Automated exchange rate forecasting	Structured pipeline (data acquisition, data processing, feature selection, and model forecasting)	End-to-end automated forecasting	Implemented via evidence-based critique
FinVerse ^[22]	Single-agent react loop	General-purpose financial analysis	Dynamic tool-use (application programming interfaces (APIs), code interpreter)	High flexibility and versatility	No hallucination solution, relying on tool output correctness
FINMEM ^[21]	Cognitive single-agent with layered memory	Autonomous stock trading	Human-like memory retrieval	Adaptability and human-like learning	No hallucination solution, relying on internal memory consistency
FinGPT ^[19]	Open-source FinLLM framework	Enhancing accessibility of FinLLMs	Lightweight adaptation (LoRA, RLSP) ^[19]	Accessibility and low-cost customization	No hallucination solution, addressed indirectly via data quality
FinLlama ^[20]	Fine-tuned foundational model	Financial sentiment classification	Task-specific classification head	High accuracy on a specific task	Not applicable (classification task)

decision, and action, each with its own limitations. At the perception stage, traditional exchange rate forecasting methods primarily rely on structured data sources, such as economic indicators or financial metrics, while often neglecting valuable unstructured information like market sentiment or policy communications. Moreover, the acquisition of such heterogeneous and multi-source data is typically handled through manual downloads or fragile, custom-built scrapers tailored to specific sources. This approach results in limited data coverage, poor timeliness, and a lack of adaptability to frequent changes in financial data APIs. These constraints fundamentally limit the forecasting capacity and generalizability of traditional models. At the planning stage, they frequently depend on manual processes, significantly restricting data processing efficiency. At the decision stage, feature selection depends on expert judgment, which increases uncertainty due to a lack of systematic application. Finally, the action stage remains heavily reliant on expert intervention for model selection and hyperparameter tuning, and also lacks automated optimization mechanisms, thereby rendering the overall process subjective, inefficient, and difficult to deploy in agent-based financial forecasting systems.

3.2 Solution based on FX-Agents paradigm

To overcome these limitations, we propose the FX-Agents framework, a multi-agent paradigm consisting of a perception agent (hereinafter referred to as PA1), planning agent (hereinafter referred to as PA2), decision agent (hereinafter referred to as DA), and

forecasting agent (hereinafter referred to as FA), as illustrated in Fig. 1. This paradigm redefines the entire workflow by assigning specialized roles to each agent, with each role designed to solve a specific limitation of the traditional process:

- **PA1 (Perception stage)** overcomes the narrow scope of traditional methods by collecting comprehensive, multimodal data from diverse sources.
- **PA2 (Planning stage)** replaces cumbersome manual workflows by automating the preprocessing and initial feature engineering of the collected data.
- **DA (Decision stage)** replaces subjective expert knowledge with dynamic, evidence-based feature selection via an RAG mechanism.
- **FA (Action stage)** constitutes the framework's final action by automating model training, optimization, and forecasting, thus overcoming the inefficiencies of manual model forecasting.

The subsequent sections detail the specific implementation of each agent within this collaborative framework.

3.3 FX-Agents: Multi-agent implementation

FX-Agents is based on LangGraph^[11], an efficient multi-agent development platform providing collaborative task scheduling, data flow management, and dynamic interaction mechanisms to enhance cooperation and adaptability among agents in exchange rate forecasting tasks.

3.3.1 PA1

The primary challenge in traditional exchange rate forecasting is the complexity and inefficiency of acquiring heterogeneous, multi-source data. To

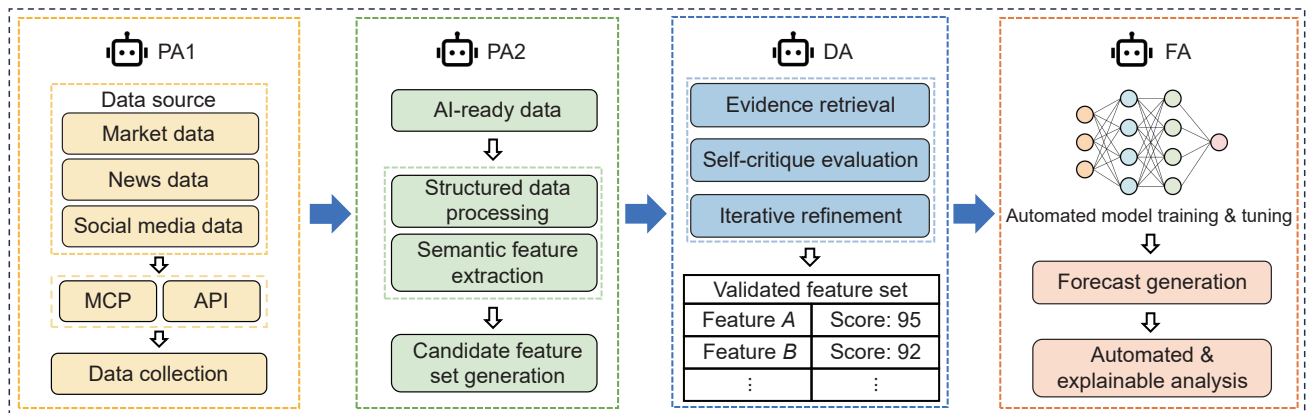


Fig. 1 FX-Agents: multi-agent collaborative workflow framework. The process begins with PA1 collecting multimodal data, which are then processed into a candidate feature set by PA2. Subsequently, DA employs an evidence-based self-critique loop to validate and select the final features. These validated features are then utilized by FA for automated model tuning and the generation of an explainable forecast.

systematically address this bottleneck, FX-Agents introduces PA1 as the framework’s automated, multimodal data perception, and acquisition module. Upon receiving a high-level task, PA1 first queries its internal tool knowledge base, a pre-constructed repository detailing the functionality and interface of each available tool, to intelligently decompose the high-level task into a series of concrete, executable data collection sub-tasks and to plan the optimal data acquisition path.

Conversely, when dealing with unstructured data sources that lack official APIs, such as real-time news or government statements, PA1 adopts the model context protocol (MCP)^[23], an open standard that governs interactions between AI models and external

tools. Under this protocol, PA1 reformulates the web scraping task as a standardized structured request, as illustrated in Fig. 2. This request is sent to the Firecrawl service, which handles the complex scraping process and returns the content in a clean, semantically rich, “AI-Ready Data^[24]” format. This design enables the agent to go beyond the traditional “crawl-and-pars” approach.

This dual-track strategy, which enables the collection of various data types as summarized in Table 2, thoroughly decouples the agent’s “reasoning core” from the “execution details” of all data sources, regardless of whether their interfaces are known, unknown, public, or private. This not only ensures the efficiency and breadth of data acquisition but also, by

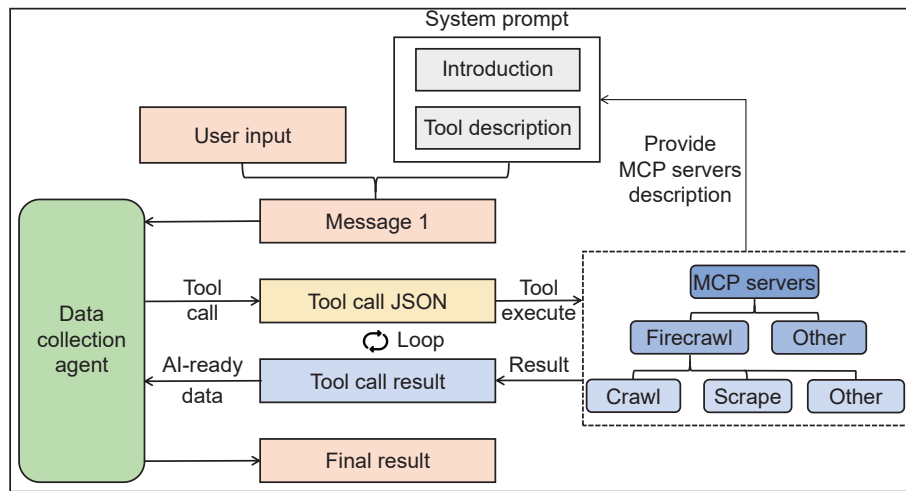


Fig. 2 PA1 data acquisition via MCP.

Table 2 PA1’s dual-track data acquisition strategy and collection types.

Category		Data name	Source	Acquisition method	Example
Track 1: Data via function calling (for stable interfaces)	Macroeconomic indicator	US federal funds rate, CN CPI	Federal Reserve Economic Data API, National Bureau of Statistics of China	Function calling (API)	“US_Fed_Rate = 5.50%”
	Financial market data	Onshore/offshore spot rate	Wind Database, Yahoo Finance	Function calling (API)	“USD/CNH_Spot = 7.2530”
	Custom policy/sentiment	User-defined event file	User upload	Function calling (local file)	“Sentiment.xlsx”
	Real-time news & commentary	Market-moving headline	Reuters, Bloomberg, Twitter	MCP + Firecrawl (Web)	“AI-Ready Text: Fed officials hint...”
Track 2: Data via MCP (for unstructured web sources)	Official government statements	Central bank announcement	People’s Bank of China (PBoC), US Department of the Treasury	MCP + Firecrawl (Web)	“AI-Ready Text: PBoC releases Q2...”
	Industry research report	Investment bank analysis	Institutional website	MCP + Firecrawl (Web)	“AI-Ready Text: Analysis on capital...”

adhering to advanced protocols like MCP, provides the entire FX-Agents framework with the robustness and extensibility necessary to tackle complex exchange rate forecasting problems.

3.3.2 PA2

PA2 is designed to address the complex and cumbersome nature of data processing in conventional forecasting. As the framework's dedicated data processing module, its workflow begins upon receiving the multimodal data collected by PA1. This handoff includes structured time-series data as well as the model-interpretable "AI-Ready Data" generated from unstructured sources, ensuring a clear and standardized data interface between the agents.

To efficiently process the incoming mix of structured and unstructured data, PA2 adopts a hybrid processing strategy driven by an LLM, the core workflow of which is detailed in Algorithm 1. For structured data, it invokes a suite of deterministic programmatic tools to execute essential cleaning and transformation tasks, such as resampling time series data to a unified daily frequency, handling missing values through interpolation techniques, and applying normalization to ensure scale consistency across variables. For semantically rich unstructured text, PA2 leverages its core LLM to perform intelligent semantic transformation. Taking the construction of a key sentiment feature as an example, to ensure the process is stable and reliable, PA2 calls a single-responsibility internal sentiment analysis tool. This tool constructs a particular prompt based on a predefined template, for instance, "Please analyze the following news text for its potential impact on the Sino-US exchange rate and return a sentiment score between -1 (highly negative) and $+1$ (highly positive)" and efficiently converts

complex textual narratives into standardized quantitative indicators.

Through this design, PA2 ultimately consolidates the various data sources into a single structured candidate feature set. This set of features is then systematically passed to the downstream DA for the subsequent evaluation and selection stage.

3.3.3 DA

Traditional feature selection, a critical stage in forecasting, often relies on opaque statistical methods or subjective expert judgment, resulting in a process that lacks transparency and is unable to dynamically incorporate new findings. To address these limitations, this paper introduces the DA, an evidence-based evaluation framework designed for intelligent feature selection. The DA receives a candidate feature set from the PA2 and systematically evaluates each feature using a framework that integrates RAG technology with an LLM. Its primary function is to select these candidates, ultimately sending a refined set of evidence-backed high-quality features to the downstream FA.

The design of DA draws considerable inspiration from the SELF-RAG framework^[25], which enhances the factual accuracy and overall quality of generated content by enabling LLMs to retrieve information on demand, generate relevant outputs and engage in critical self-reflection. This study introduces an innovative adaptation of this concept within the domain of feature engineering for exchange rate forecasting. In this context, DA functions as a seasoned macroeconomic analyst. It receives a set of candidate features from the upstream PA2 and conducts evidence-based evaluations to assess their relevance and predictive utility. Following this rigorous assessment process, DA delivers a well-validated, high-quality feature set with strong forecasting potential to the FA. To fulfill this role, DA operates through a three-stage workflow encompassing evidence retrieval, self-critical analysis, and iterative refinement. The overall workflow of the DA is illustrated in Fig. 3.

Step 1: Evidence retrieval

All decisions made by the DA are grounded in robust, evidence-based reasoning. To address common issues in LLMs, such as hallucinations and over-reliance on unverified online information, this paper introduces a domain-specific knowledge base in financial and macroeconomic analysis tailored for

Algorithm 1 PA2 workflow

Input: Raw data D_{raw} from PA1 and tool set
Output: Candidate feature set D_{feature}
1: Initialize LangGraph environment;
2: Load tool set;
3: Obtain raw data D_{raw} ;
4: **for** each data $d \in D_{\text{raw}}$ and d is unstructured **do**
5: Construct prompt;
6: $d_{\text{structured}} \leftarrow \text{PA2.use.tool (with prompt)}$;
7: **end**
8: **for** each structured data d in $D_{\text{structured}}$ **do**
9: $d_{\text{clean}} \leftarrow \text{PA2.use.tool (process structured data)}$;
10: **end**
11: $D_{\text{feature}} \leftarrow \text{PA2.use.tool (analyze data, construct features)}$;
12: **return** D_{feature}

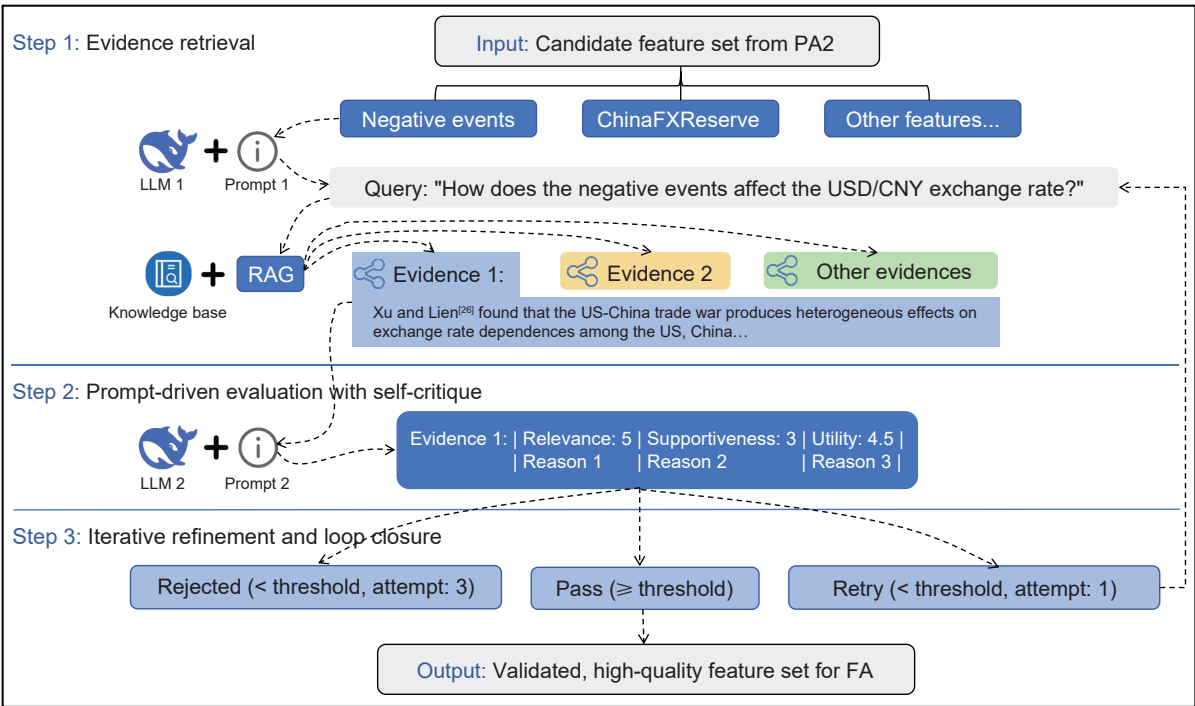


Fig. 3 DA workflow overview.

exchange rate forecasting. This knowledge base is constructed from carefully curated and pre-processed authoritative sources, including:

- **Academic literature:** Peer-reviewed journal articles from major databases in economics and finance, particularly in areas such as exchange rates and monetary policy, ensuring theoretical rigor and academic credibility.
- **Official reports and data:** Publications and statistical datasets from central banks of major economies (e.g., the Federal Reserve System, the People’s Bank of China), as well as international institutions such as the International Monetary Fund and the Bank for International Settlements, offering authoritative monetary policy reports and economic outlooks.

• **Trusted financial media:** Historical reports and in-depth analyses from reliable media sources such as Reuters and Bloomberg, providing contextual insights into key financial events.

During candidate feature selection, the DA performs evidence retrieval through the following steps:

(1) **Query generation:** When the DA receives a candidate feature (e.g., US_1Y_Treasury_Yield), it first reformulates the feature into a more specific natural language question, for instance: “How does the change in the 1-year US Treasury yield affect the

USD/CNY exchange rate?”

(2) **Vectorized retrieval:** The generated question is converted into a high-dimensional vector representation, which is then used to perform a semantic similarity search within the knowledge base’s vector index.

(3) **Content chunking:** The retrieval system returns the text fragments (chunks) that are most relevant to the query. These fragments, along with their metadata (such as source and publication date), are submitted to the DA as direct evidence for the subsequent evaluation stage.

Step 2: Prompt-driven evaluation with self-critique

To prevent the DA from introducing noise when the retrieved evidence is unreliable, a structured prompt is used after evidence retrieval to guide the LLM in conducting a critical evaluation. Drawing on the design principles of self-RAG, this paper guides the LLM to produce a series of specialized critique tokens. These tokens decompose the overall judgment into three dimensions: relevance, supportiveness, and utility, as summarized in Table 3. For each dimension, the model generates both a quantitative score and a qualitative justification, thereby making the evaluation process transparent and interpretable.

After scoring each critique dimension individually,

Table 3 Evaluation dimensions of critique tokens for the validation module.

Dimension	Score	Description
Relevance	[0, 5]	Assess the relevance of the generated argument to the forecasting task
Supportiveness	[0, 5]	Evaluate how well the evidence supports the generated argument
Utility	[0, 5]	Measure the feature’s potential contribution to model performance based on the argument

the DA proceeds to compute an overall score for each feature through a two-step process. First, the raw composite score S_{raw} is calculated by Eq. (1),

$$S_{\text{raw}} = \sum_{G \in \{\text{critique tokens}\}} W_G S_G \quad (1)$$

where S_G denotes the score assigned to each critique dimension (ranging from 0 to 5), and W_G is the weight configured by the user for each dimension.

Next, to allow intuitive comparison across features, the raw scores are linearly normalized to the interval [0, 100] in Eq. (2),

$$S_{\text{normalized}}(x) = 100 \times \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (2)$$

where x represents the raw composite score S_{raw} computed for each feature using Eq. (1). Subsequently, $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of these raw scores across all candidate features, respectively. Final feature selection is based on whether the normalized score exceeds a user-defined threshold.

Step 3: Iterative refinement and loop closure

To avoid wrong judgments caused by incomplete evidence retrieval, the DA is equipped with an iterative refinement and loop closure mechanism. When a candidate feature receives a low initial evaluation score, the DA does not immediately discard it, but instead initiates a reflection and optimization cycle. It first analyzes the specific reasons behind the low score, which may include missing key evidence in the initial retrieval, weak support from the retrieved documents, or bias in the evaluation perspective. Based on this reflection, the DA formulates a more targeted and precise query, initiating a second round of retrieval to obtain more relevant or stronger evidence. It then re-evaluates the newly retrieved content using the same multi-dimensional critique process. Suppose the new score still fails to meet the required threshold. In that case, the DA continues this refinement loop until the composite score exceeds the predefined cutoff or the maximum number of allowed iterations is reached, thereby ensuring that the feature selection process remains rigorous yet flexible.

For instance, when evaluating the feature “negative events”, the DA initially generates the query: “How do negative events related to the Sino-US trade war affect the USD/CNY exchange rate?” However, the retrieved evidence is insufficient to support the feature. The DA then triggers a reflection step, recognizing that the initial query lacked specificity regarding economic mechanisms and policy responses. It refines the query to: “How do Sino-US trade tensions, along with central bank policy reactions, influence CNY exchange rate volatility?” This revised query leads to the retrieval of more targeted and relevant literature. With stronger support, the feature is ultimately accepted into the high-quality feature set.

Through this workflow, the DA effectively leverages the retrieval-augmented generation and self-critique strengths of self-RAG, reducing hallucinations and improving decision quality. This ensures that the feature selection process remains scientific, transparent, and interpretable. Ultimately, the high-quality features selected by the DA significantly enhance the performance of the FA, improving both the accuracy and stability of the overall framework.

3.3.4 FA

The FA within the FX-Agents framework plays a central role in generating final forecasts and conducting performance evaluations. Initially, the FA receives a set of pre-selected features from the DA and refines them further to suit the forecasting task. Although powerful automated hyperparameter optimization frameworks like Optuna^[27] exist, their application in traditional financial forecasting workflows often remains isolated and fragmented. The key challenge lies not in the absence of such tools but in the lack of end-to-end frameworks that seamlessly integrate dynamic feature selection, model optimization, and performance evaluation.

The FA is explicitly designed to address this integration gap. It does not merely act as a wrapper for Optuna, it serves as the final orchestrator in the agent pipeline. Upon receiving a set of evidence-based features from the DA, the FA leverages the embedded Optuna framework to immediately initiate an efficient hyperparameter search, which utilizes heuristic

strategies such as Bayesian optimization. This approach facilitates the rapid discovery of optimal or near-optimal configurations for forecasting models such as recurrent neural networks (RNNs), LSTMs, and Transformers, thereby improving predictive accuracy on the target dataset.

Once the optimal model and its hyperparameters are determined, the FA initiates a complete training procedure to generate exchange rate forecasts. To enhance the interpretability and transparency of the forecasting process, the FA integrates model explanation techniques such as shapley additive explanations (SHAP)^[28], which quantitatively assesses the contribution of individual features and visually presents their impacts through feature importance and partial dependence plots. Furthermore, by leveraging LLMs' natural language generation capabilities, these visual insights are translated into accessible analytical reports. These reports clearly articulate the key drivers behind exchange rate fluctuations and elucidate the specific roles played by each input feature in the model's predictions. As a result, decision-makers are equipped with accurate, efficient, and highly interpretable forecasting outputs.

Overall, FX-Agents significantly improves the comprehensiveness and accuracy of exchange rate forecasting by coordinating data collection, preprocessing, feature selection, and forecasting through a collaborative multi-agent paradigm. The following section systematically evaluates the performance of FX-Agents across various data scenarios.

4 Experiment

Accurate and stable exchange rate forecasting faces challenges in efficiently collecting and processing big data while extracting key forecasting features. This paper proposes the FX-Agents framework to address these challenges, with its feasibility validated through three research questions:

- Does the multi-agent workflow in FX-Agents collaborate to improve the prediction performance of forecasting models (Section 4.3)?
- Compared to the features selected by traditional feature engineering, do the features selected by FX-Agents demonstrate superior performance across different forecasting models (Section 4.4)?
- Does FX-Agents exhibit higher data processing efficiency compared to traditional feature engineering (Section 4.5)?

4.1 Experimental data and setup

To address the three research questions, this paper employs the USD/CNY exchange rate as the forecasting target to evaluate the performance and efficiency of FX-Agents in exchange rate forecasting.

Sample period. This study analyzes the USD/CNY exchange rate from January 1, 2017, to December 31, 2024. The period witnessed heightened exchange rate volatility due to escalating Sino-US trade tensions, which facilitates the evaluation of the capability of FX-Agents in capturing exchange rate volatility.

Data sources. The experimental data were collected through the PA1 within the FX-Agents framework. The PA1 retrieved time-series data through the API of the Wind Database and automated the collection of trade-related policy and news event data from official government sources, including the State Council Information Office of the People's Republic of China, the Ministry of Commerce of the People's Republic of China, the US Department of State, and the US Department of Commerce.

Benchmark models and setup. This paper selects mainstream exchange rate forecasting models as benchmarks, including the state-of-the-art (SOTA) Transformer model, alongside other representative models like LSTM + Attention, RNN^[29], temporal fusion Transformer (TFT)^[30], and TimesNet^[31]. All benchmark models are trained with a standardized set of core hyperparameters: a hidden dimension of 64, a dropout rate of 0.1, a learning rate of 0.001, and a training duration of 300 epochs. Additionally, the FX-Agents leverages the DeepSeek-R1 (671B) model as the base LLM for intelligent task execution. To mitigate the computational demands of local deployment, we utilized its official API version.

Evaluation metrics. This paper employs root mean square error (RMSE) and mean absolute percentage error (MAPE) as evaluation metrics^[6]. RMSE measures the absolute deviation between predicted and actual values, while MAPE assesses relative forecasting accuracy. Lower values for both metrics indicate superior model performance.

Hardware configuration. All experiments in this paper are conducted on a Windows 11 system with the following computational environment: Python 3.10.16, PyTorch 2.1.2, and CUDA 11.8. The hardware configuration consisted of an Intel Xeon Platinum 8352V CPU and an NVIDIA RTX 4080 GPU.

4.2 Feature engineering based on FX-Agents

The FX-Agents framework is designed to address key challenges in exchange rate forecasting, including the complexity of multimodal data integration, inefficient big data processing, and non-standardized feature selection. Accordingly, this experiment conducted feature engineering through the multi-agent workflow of the FX-Agents framework.

Time-series feature construction. Feature selection is crucial for exchange rate forecasting, particularly when involving macroeconomic indicators and relevant features^[32]. To improve forecasting accuracy for USD/CNY, time-series features directly correlated with USD/CNY are essential, allowing the model to capture exchange rate volatility better. In this experiment, the FX-Agents framework retrieves relevant time-series features from both Google Scholar and a local knowledge base. The PA1 then collects the data, while the PA2 ensures its integrity and temporal consistency. These time-series features are listed in Table 4.

Event sentiment feature construction. Trade conflicts, policy document releases, and bilateral negotiations between China and the United States have a significant influence on exchange rate fluctuations^[6]. To improve the forecasting model's ability to capture such volatility, this paper develops event sentiment features through a multi-agent workflow within the FX-Agents framework. The process involves four main stages. (1) The PA1 automatically collects 1030 policy and news events related to Sino-US interactions during the sample period from official government sources, using predefined keywords such as “China”, “United States”, and “Sino-US”. (2) The PA2 filters the initial dataset by eliminating events unrelated to bilateral trade, such as those involving military affairs, geopolitical conflicts, or third-party countries, resulting

in 429 trade-relevant events. (3) The PA2 conducts event type classification and sentiment analysis for these filtered events through API integration with the DeepSeek-R1 model. (4) The PA2 transforms the classified events into one-hot encoding, where a value of 1 represents event occurrence on a specific date and 0 otherwise. For events occurring on holidays, the corresponding one-hot encoding is reassigned to the next available trading day to ensure temporal alignment with USD/CNY exchange rate data. The final event sentiment features constructed by FX-Agents, along with their sample size, are summarized in Table 5. Additionally, Fig. 4 is plotted to provide a preliminary observation of the relationship between these event sentiment features and exchange rate fluctuations.

Figure 4 illustrates a clear correlation between event sentiment features and exchange rate fluctuations. Specifically, during periods with a high frequency of negative events, the CNY tends to depreciate noticeably.

Features selection. After completing the construction of time-series features and event sentiment features, the DA conducts a comprehensive evaluation of all features and executes the final feature selection through the following workflow: (1) The DA first performs evaluation analyses based on the features and retrieved feature-related research materials from policy, industry, and academic knowledge bases to establish the foundation for feature selection. (2) The DA evaluates each retrieved research material with a relevance score, indicating the degree of relevance between the material and the features. (3) The DA evaluates each material with a supportiveness score, assessing how supportive the material is for feature selection. (4) Finally, the DA evaluates the selected results (containing outputs from LLMs and externally retrieved evidence) with a utility score, evaluating the

Table 4 Time series feature information table.

Feature name	Feature meaning	Feature sample
Date	Date	2017-02-28
USD/CNY	USD/CNY exchange rate	6.8689
USD_Index	US Dollar Index	101.3437
CN_1Y_GovBond_Yield	China 1-year government bond yield	2.72%
US_1Y_Treasury_Yield	US 1-year Treasury bond yield	0.88%
China_FX_Reserves	China's foreign exchange reserves	3.005 trillion USD
CN_CPI_MoM	China consumer price index month-over-month change	0.8%
US_CPI_MoM	US consumer price index month-over-month change	2.2%
WTI_Futures_Price	West Texas Intermediate crude oil futures settlement price	54.01 USD

Table 5 Event sentiment features and their corresponding sample sizes, sorted by sample size in descending order, with some intermediate features omitted.

Feature	Sample size
Negative events	248
Negative events from the US	215
Positive events	181
US restrictions on Chinese enterprises	160
Sino-US dialogue	114
Negative events from China	45
Positive events from the US	39
Positive events from China	34
⋮	⋮
China increasing purchases of US goods	4
China reducing restrictions on US enterprises	3
US designating China as currency manipulator	1
US removal China from currency manipulator list	1

usefulness of the results for feature selection. All three metrics, relevance, supportiveness, and utility, employ a unified scoring range of $[0, 5]$, and the comprehensive score is calculated as Eq. (3),

$$S(\text{feature}) = S_{\text{normalized}}(0.2 \times \text{relevance} + 0.4 \times \text{supportiveness} + 0.4 \times \text{utility}) \quad (3)$$

where $S(\text{feature})$ denotes the comprehensive score of the feature. In this experiment, the weights (W_G) for the three metrics in Eq. (3) are calibrated through a grid search experiment on an independent validation set. We test multiple weight combinations and find that setting the weights for “supportiveness” and “utility” to 0.4, and for “relevance” to 0.2, achieves optimal forecasting performance. Additionally, to make this composite score intuitively comparable across different features, we map it to the standard $[0, 100]$ interval

using the linear normalization function $S_{\text{normalized}}(x)$, as shown in Eq. (2).

After completing the scoring process, the DA ranks all features in descending order of their scores and selects features with composite scores meeting or exceeding a threshold of 80 as the final selected feature set. These high-scoring features are identified as exhibiting significant correlation with USD/CNY exchange rates and are validated through multi-source evidentiary support. As shown in Table 6, “negative events” receives the highest score because DA finds sufficient evidence to support its impact on the exchange rate. Notably, this feature aligns with the pattern observed in Fig. 4, where frequent negative events impact the exchange rate, highlighting the robustness of the features selected by the DA. Conversely, “US removal of China from currency manipulator list” scores the lowest due to a lack of sufficient supporting evidence. This result demonstrates how the evidence-based nature of the framework enhances the traceability and reliability of its conclusions, reflecting the strength of our framework in addressing the limitations of the LLMs hallucination. Furthermore, this evidence-driven feature selection method offers extensibility. By updating the policy, industry and academic knowledge bases, the method can meet the feature selection requirements for forecasting exchange rates between different currency pairs.

4.3 Ablation experiments on FX-Agents multi-agent workflow

This section conducts an ablation study to validate the contribution of each agent within the FX-Agents workflow. We measure this by observing the

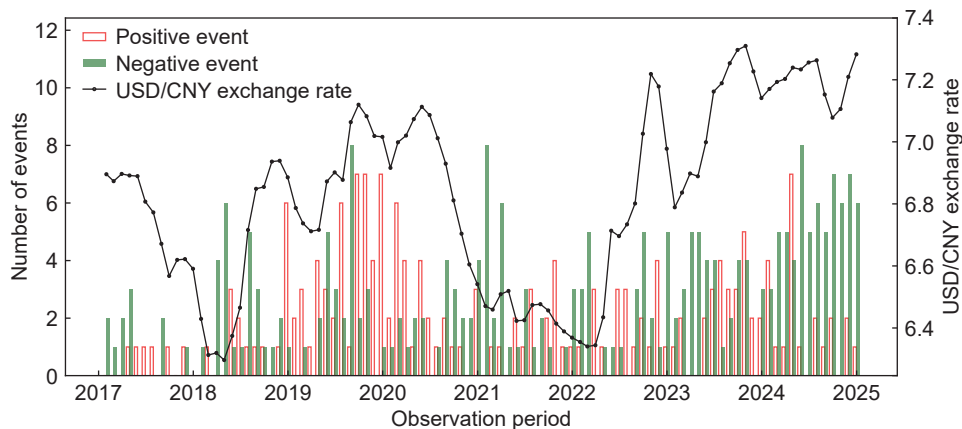
**Fig. 4** Events and exchange rate fluctuations with each bar representing the monthly count of positive or negative events.

Table 6 DA-based feature selection with evaluation scores.

Feature	DA interpretation and justification	Relevance score	Supportiveness score	Utility score	Evaluation score
Negative events	The trade war produces heterogeneous effects on exchange rate dependences among US, China, and their major trade partners, further influencing the decisions of portfolio diversification, risk management, central bank interventions, and international trade ^[26] . Han et al.'s (2024) findings suggest that prioritizing the collection of negative-type events as critical features in forecasting frameworks can enhance exchange rate forecasting accuracy ^[6] .	5.0	5.0	5.0	100
China_FX_Reserves	China's central bank can manage exchange rate volatility by adjusting foreign exchange reserves. Ample reserves indicate strong intervention capacity and reduce risk of sharp depreciation ^[33] .	5.0	4.0	5.0	92
WTI_Futures_Price	The comovement between exchange rates and oil prices is time varying in nature. The shocks from oil volatility on four CNY exchange rates are strong and persistent ^[34] .	4.5	4.0	5.0	90
US_1Y_Treasury_Yield	Higher US yields attract capital inflows to dollar assets, causing USD appreciation and CNY depreciation. This is consistent with the covered interest parity (CIP) theory ^[35] .	5.0	3.0	4.5	80
⋮					
US Removal of China from Currency Manipulator List	This symbolic policy offers short-term optimism but lacks fundamental impact on long-term exchange rates.	0	0	1.0	8

Note: The ellipsis indicates omitted candidate features with intermediate DA evaluation scores.

performance of the forecasting agent, denoted as $FA(\chi)$, where χ is the encapsulated model (e.g., Transformer and RNN). The results in Table 7 demonstrate a progressive performance enhancement for $FA(\chi)$ as the PA1, PA2, and DA are sequentially introduced.

According to the process of feature engineering, PA1 is responsible for collecting time-series features, PA2 for generating event sentiment features, and DA for feature selection. Consequently, in this experiment, the feature input for “ $FA(\chi) + PA1$ ” includes all time-series features, while “ $FA(\chi) + PA1 + PA2$ ” incorporates all time-series features combined with event sentiment features. The feature input for “ $FA(\chi) + PA1 + PA2 + DA$ ” consists of the features selected by

DA, allowing us to evaluate the contribution of its LLM-based feature selection.

Based on the results in Table 7, it is evident that the multi-agent workflow of FX-Agents significantly improves the performance of forecasting models. For instance, expanding the framework from “ $FA(\text{Transformer}) + PA1$ ” to “ $FA(\text{Transformer}) + PA1 + PA2 + DA$ ” leads to a significant improvement in forecasting performance, with the RMSE decreasing from 0.0936 to 0.0566 and the MAPE decreasing from 1.1220% to 0.6532%. Moreover, the experimental results indicate that the performance of forecasting models improves progressively with the introduction of PA1, PA2, and DA, achieving the best performance under the “ $FA(\chi) + PA1 + PA2 + DA$ ” configuration.

Table 7 Ablation experiments on FX-Agents multi-agent workflow (values in bold indicate the best performance, values with underlines indicate the second-best performance).

Model	FA(χ) + PA1		FA(χ) + PA1 + PA2		FA(χ) + PA1 + PA2 + DA	
	RMSE	MAPE (%)	RMSE	MAPE (%)	RMSE	MAPE (%)
Transformer	0.0936	1.1220	<u>0.0737</u>	<u>0.8739</u>	0.0566	0.6532
LSTM + Attention	0.0919	1.1492	<u>0.0876</u>	<u>1.0862</u>	0.0599	0.7205
RNN	0.0791	0.9710	<u>0.0762</u>	<u>0.9529</u>	0.0650	0.8121
TFT	0.0383	0.4540	<u>0.0352</u>	<u>0.3693</u>	0.0330	0.3427
TimesNet	0.0829	0.9858	<u>0.0715</u>	<u>0.8362</u>	0.0436	0.4874

4.4 Process and performance comparison: FX-Agents vs. traditional feature engineering

To evaluate the performance of features selected by FX-Agents versus traditional feature engineering across different forecasting models, this experiment implements the traditional feature engineering process as follows.

Time-series feature selection. To identify time-series features with high correlation to USD/CNY, this experiment employs Pearson correlation coefficients for feature selection^[36]. Since China_FX_Reserves, CN_CPI_MoM, and US_CPI_MoM are monthly frequency indicators, they require temporal alignment to daily frequency data matching USD/CNY for correlation computation. Because these features have different economic effects, they require distinct preprocessing methods: (1) As a stock variable demonstrating relative stability between months, China_FX_Reserves adopts forward-fill imputation (carrying the last observed value forward) for missing values. (2) Since CN_CPI_MoM and US_CPI_MoM generally exhibit stable trends, linear interpolation (estimating intermediate values through straight-line connections between existing data points) is employed to connect CPI values across different months, thereby avoiding excessive volatility.

After temporal alignment, all time-series variables are standardized to normalize their scales, reducing potential bias in Pearson correlation coefficients due to magnitude differences.

As shown in Fig. 5, CN_1Y_GovBond_Yield, USD_Index, and US_1Y_Treasury_Yield exhibit strong correlations with USD/CNY. However, incorporating all three features risked information redundancy. Given the observed correlation (>0.7) between USD_Index and US_1Y_Treasury_Yield, we retain only USD_Index due to its higher correlation with USD/CNY. The final selected time-series features are USD_Index and CN_1Y_GovBond_Yield.

Event sentiment features selection. Since event sentiment features are dummy variables, their Pearson correlation with USD/CNY could not be computed directly. Therefore, this experiment employs the XGBoost algorithm to calculate importance weights for each event sentiment feature^[37]. The calculation formula is defined by Eq. (4),

$$\text{Weight}(f) = \sum_{t=1}^T \sum_{i=1}^N 1(f_i = \text{split}_t) \quad (4)$$

where T is the total number of trees in the model and N denotes the number of features. The indicator function $f_i = \text{split}_t$ equals 1 if feature f_i is selected as a splitting variable at the t -th tree, and 0 otherwise. This formulation calculates how frequently each feature is chosen for splits in the decision trees. A higher frequency indicates greater forecasting importance within the model structure.

Figure 6 displays the importance weights of all event sentiment features. We apply a mean-threshold filter,

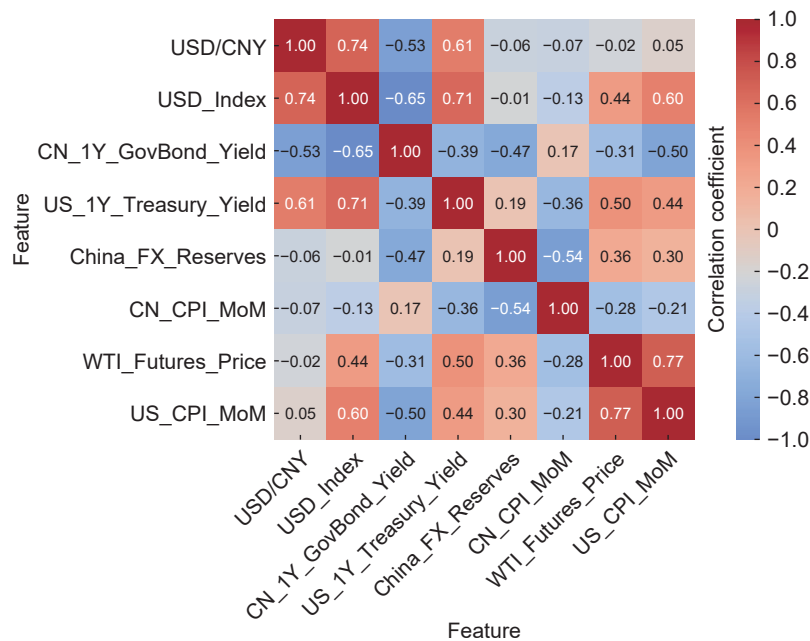


Fig. 5 Time-series feature correlation coefficient heat map.

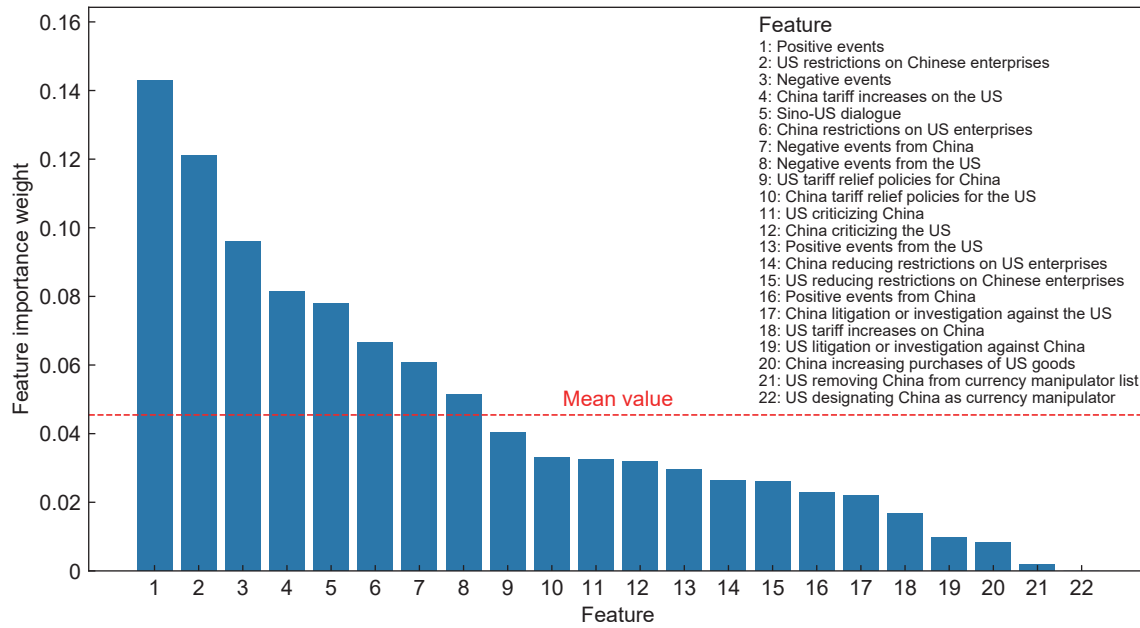


Fig. 6 Feature importance weight.

retaining only features with above-average weights. Due to information overlap (e.g., Sino-US dialogue within positive events, and US restrictions on Chinese enterprises within negative events), only two features are finally retained: positive events and negative events.

Comparative experiments. The features selected by FX-Agents and traditional feature engineering are shown as Table 8.

The two feature sets are independently fed into distinct forecasting models for comparative analysis. The results are summarized in Table 9.

As evidenced in Table 9, features selected by FX-Agents demonstrate more significant performance improvements across multiple forecasting models (Transformer, TFT, TimesNet), suggesting stronger generalization capability and robust forecasting performance. Notably, traditional models, such as RNN and LSTM + Attention, experience a slight

Table 8 Comparison of features selected by FX-Agents and traditional feature engineering.

Feature selected by FX-Agents	Feature selected by traditional feature engineering
Negative events	USD_Index
China_FX_Reserves	CN_1Y_GovBond_Yield
WTI_Futures_Price	Positive events
US_1Y_Treasury_Yield	Negative events

performance degradation when using features selected by FX-Agents. This is because traditional models are adept at capturing stable time-series features, whereas the features selected by FX-Agents involve more complex non-linear relationships that require models with greater expressive power to be fully exploited^[38]. This also explains why in mainstream models (e.g., Transformer) the features from FX-Agents lead to more significant performance improvements. The combined results from Tables 7 and 9 fully

Table 9 Comparison of forecasting performance between FX-Agents and traditional feature engineering. Values in bold indicate the best performance. “+” indicates that the features selected by FX-Agents improve model performance compared to traditional feature engineering, while “-” indicates a decline.

Forecasting model	FX-Agents		Traditional feature engineering		Performance difference	
	RMSE	MAPE (%)	RMSE	MAPE (%)	RMSE	MAPE (%)
Transformer	0.0566	0.6532	0.0758	0.8991	+25.3298	+27.3496
LSTM + Attention	0.0599	0.7205	0.0556	0.6607	-7.7338	-9.0510
RNN	0.0650	0.8121	0.0598	0.7348	-8.6956	-10.5199
TFT	0.0330	0.3427	0.0365	0.3915	+9.5890	+12.4649
TimesNet	0.0436	0.4874	0.0476	0.5393	+8.4034	+9.6236

demonstrate the overall effectiveness of the FX-Agents framework and the indispensability of each of its agents.

4.5 Efficiency comparison: FX-Agents vs. traditional feature engineering

As demonstrated in Fig. 7, FX-Agents automate and intelligently execute the entire workflow, from data collection and processing to feature selection and exchange rate forecasting, through four specialized agents. In contrast to traditional feature engineering that requires heavy manual intervention, FX-Agents completes all tasks through natural language interaction, significantly enhancing operational efficiency.

Specifically, FX-Agents improves data processing efficiency by over 300 min compared with traditional feature engineering. The PA1 within FX-Agents collects nine time series and 1030 policy and news data points in 20 min, significantly faster than the minimum 180 min required for manual collection in traditional feature engineering. In the data processing phase, the PA2 efficiently cleans data, unifies temporal frequencies, and constructs event sentiment features within 20 min, whereas traditional feature engineering demands manual data cleaning and coding for sentiment analysis, consuming at least 120 min. For feature selection, the DA in FX-Agents selects features and leverages an external knowledge base in just 1 min, compared to 60 min needed for manual algorithm design and experience-based selection in traditional engineering. Finally, the FA trains five

exchange rate forecasting models in 5 min, while traditional feature engineering necessitates manual algorithm coding and parameter tuning, taking 20 min for the same task.

In summary, FX-Agents significantly enhances efficiency across data collection, processing, feature selection, and exchange rate forecasting by reducing time costs and increasing automation.

5 Conclusion and Future Work

This paper proposes an exchange rate forecasting paradigm based on multimodal big data. The paradigm achieves efficient and stable forecasting performance through standardized data processing and feature selection procedures. Based on this paradigm, we design the FX-Agents framework to overcome limitations of traditional feature engineering methods, such as low processing efficiency and non-standardized feature selection. By incorporating specialized agents (PA1, PA2, DA, and FA), the framework systematically integrates heterogeneous data, automates data cleaning and feature engineering, and performs feature selection through dynamic strategies. Experimental results demonstrate that FX-Agents outperform traditional methods in both forecasting accuracy and processing efficiency. The multi-agent collaborative mechanism significantly enhances the overall efficiency of big data processing. Furthermore, by constructing event sentiment features and applying dynamic feature selection strategies, the framework achieves superior forecasting performance.

Future research will focus on several key areas. First,

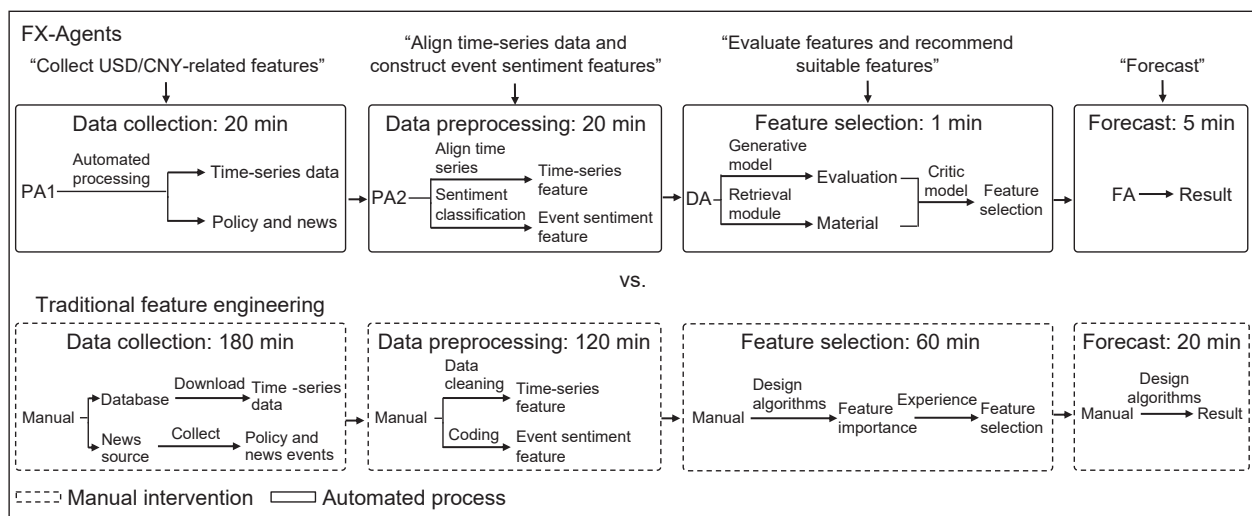


Fig. 7 Workflow comparison: FX-Agents vs. traditional feature engineering.

we will enhance the real-time forecasting capabilities of FX-Agents by integrating cross-market big data and other heterogeneous data streams to improve predictive accuracy. Second, we will investigate practical considerations, such as the API latency and computational overhead of large-scale models. Finally, we plan to explore the application of FX-Agents to other major currency pairs.

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Declaration of Competing Interest

The authors have no competing interests to declare that are relevant to the content of this article.

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